Temozolomide non-completion and absence of residual enhancement are associated with distant recurrence in IDH-wildtype glioblastoma

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OBJECTIVES Glioblastoma (GBM) recurrence pattern determines subsequent management: local recurrence may favour re-resection or re-irradiation, while distant recurrence directs systemic therapy and trial enrolment. Clinicians lack validated tools to anticipate recurrence pattern using early post-operative and treatment-course variables. We used a retrospective single-centre registry to identify early predictors of local versus distant recurrence.

METHODS Recurrence pattern (at first evidence of disease progression) was classified as local (contiguous with resection cavity) or distant on follow-up neuroimaging. A two-predictor multivariable logistic regression model was constructed, limited to two predictors given the number of distant-recurrence events to minimise overfitting, with complete-case analysis applied for missing predictor data. Discrimination was assessed by area under the receiver operating characteristic curve (AUC).

RESULTS Sixty-three IDH-wildtype patients with confirmed recurrence classification were included (local n=45, distant n=18). On multivariable analysis (complete cases, n=55), temozolomide non-completion (OR 12.4, 95% CI 1.7–88.7, p=0.012) and absence of residual contrast enhancement on early post-operative MRI (OR 4.9, 95% CI 1.2–20.4, p=0.030) were associated with distant recurrence (AUC 0.773).

CONCLUSIONS The absence of residual contrast enhancement within the resection cavity was associated with distant disease recurrence in our population. Temozolomide non-completion may reflect treatment intolerance or early occult progression. Both variables are routinely available in clinical practice before evidence of disease progression, and define a clinically tractable window for future work in early risk stratification towards a validated clinical decision tool capable of predicting recurrence patterns.
